Machine Learning Topics: Ensemble Learning

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Some slides adapted from David Sontag

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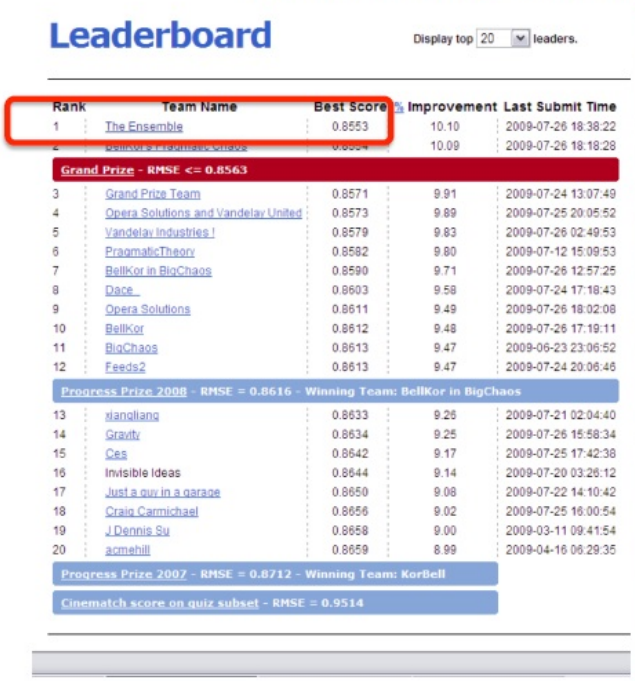
Topics

1. Motivating Example
2. Why Combine Classifiers?
3. Majority
4. Bagging
5. Stacking
6. Boosting
7. Mandatory Reading

A Contest

Ensemble methods

Machine learning competition with a $1 million prize



Leaderboard

Display top 20 leaders.

Rank	Team Name	Best Score	Improvement	Last Submit Time
1	The Ensemble	0.8553	10.10	2009-07-26 18:38:22
2	BellKor's Pragmatic Chaos	0.8554	10.09	2009-07-26 18:18:28
Grand Prize - RMSE <= 0.8563				
3	Grand Prize Team	0.8571	9.91	2009-07-24 13:07:49
4	Opera Solutions and Vandelay United	0.8573	9.89	2009-07-25 20:05:52
5	Vandelay Industries !	0.8579	9.83	2009-07-26 02:49:53
6	PragmaticTheory	0.8582	9.80	2009-07-12 15:09:53
7	BellKor in BigChaos	0.8590	9.71	2009-07-26 12:57:25
8	Dace_	0.8603	9.58	2009-07-24 17:18:43
9	Opera Solutions	0.8611	9.49	2009-07-26 18:02:08
10	BellKor	0.8612	9.48	2009-07-26 17:19:11
11	BigChaos	0.8613	9.47	2009-06-23 23:06:52
12	Feeds2	0.8613	9.47	2009-07-24 20:06:46
Progress Prize 2008 - RMSE = 0.8616 - Winning Team: BellKor in BigChaos				
13	xiangliang	0.8633	9.26	2009-07-21 02:04:40
14	Gravity	0.8634	9.25	2009-07-26 15:58:34
15	Ces	0.8642	9.17	2009-07-25 17:42:38
16	Invisible Ideas	0.8644	9.14	2009-07-20 03:26:12
17	Just a guy in a garage	0.8650	9.08	2009-07-22 14:10:42
18	Craig Carmichael	0.8656	9.02	2009-07-25 16:00:54
19	J Dennis Su	0.8658	9.00	2009-03-11 09:41:54
20	acmehill	0.8659	8.99	2009-04-16 06:29:35
Progress Prize 2007 - RMSE = 0.8712 - Winning Team: KorBell				
Cinematch score on quiz subset - RMSE = 0.9514				

users

movies					
1		?	3	5	?
?	1				2
	4		4	5	?

Intuition

Recall the Bias-Variance tradeoff[2]
Can we reduce the variance without increasing the bias?
Averaging reduces variance, when predictions are **independent**

$$Var(\overline{X}) \qquad \frac{Var(X)}{N}$$

Average models to reduce model variance
One problem:

- Only one training dataset
- Where do multiple models come from?

[2] Incidentally, bias actually denotes *bias error*

Majority

Assume binary classification:
An odd number K of classifiers are trained *on different subsets of features*
Each classifier has accuracy $.5 + \epsilon$, where $\epsilon > 0$
We take the majority vote of the K classifiers. This works well when:

- Classifiers are independent, or
- Negatively Correlated[3]

[3] Ludmila et al.,"Measures of Diversity ...," Machine Learning, 2003

Bagging: Boostrap Aggregation

- Leo Breiman (1994)
- Take repeated bootstrap samples from training set D
- *Bootstrap sampling*: Given set D containing N training examples, create D' by drawing N examples at random with replacement from D.

Bagging

1. Create k bootstrap samples $D_1 \ldots D_k$
2. Train distinct classifier on each D_i
3. Classify new instance by majority vote / average.

Random Forest Classifier - A bag of Decision Trees. Each decision tree has a random subset of the features.

Bagging

Best Case:

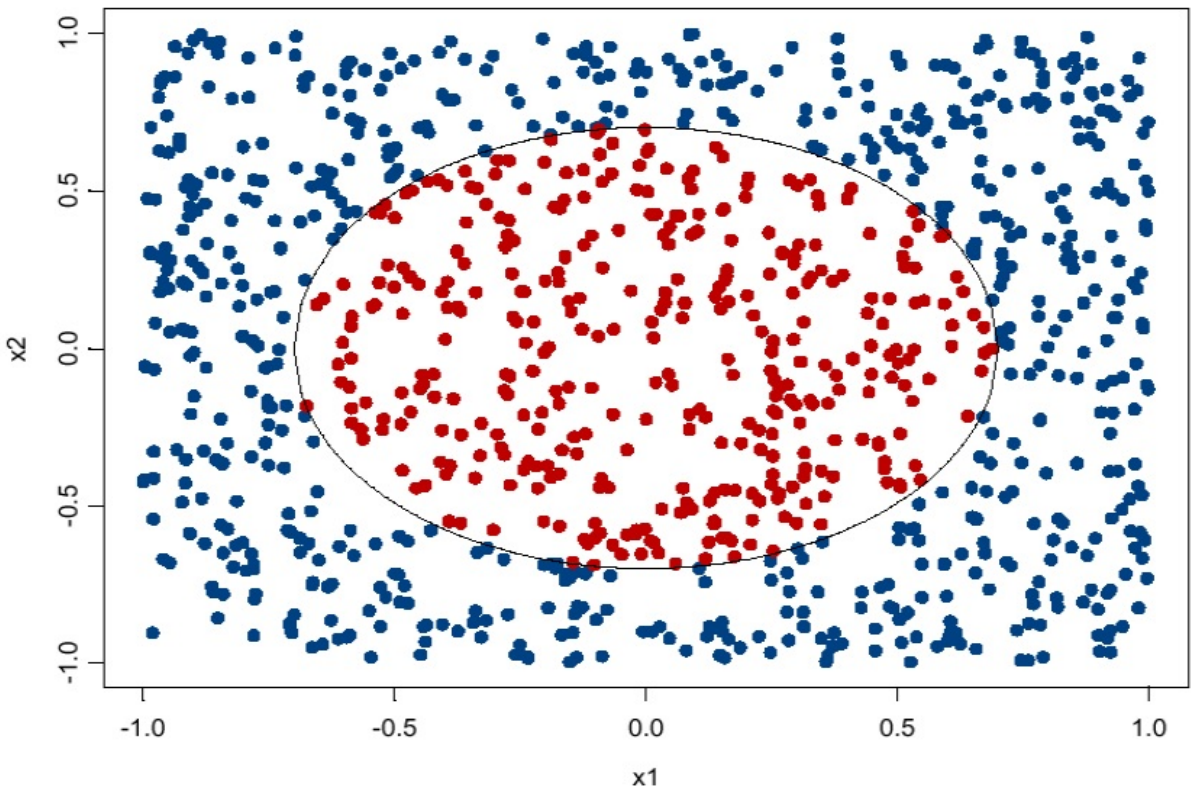
$$Var(Bagging(L(x, D))) \qquad Variance(L(x, D))/k$$

In practice:

1. Models are correlated, so reduction is smaller than $1/k$
2. Variance of models trained on fewer training cases usually somewhat larger

Example

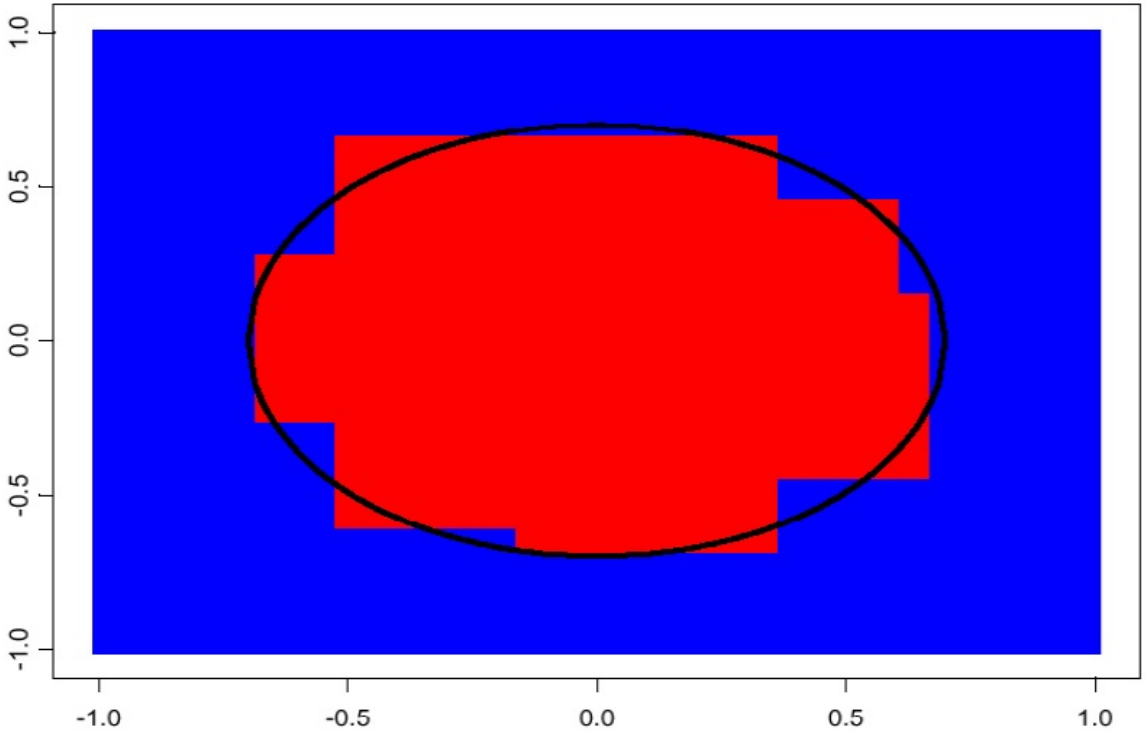
Bagging Example



Example

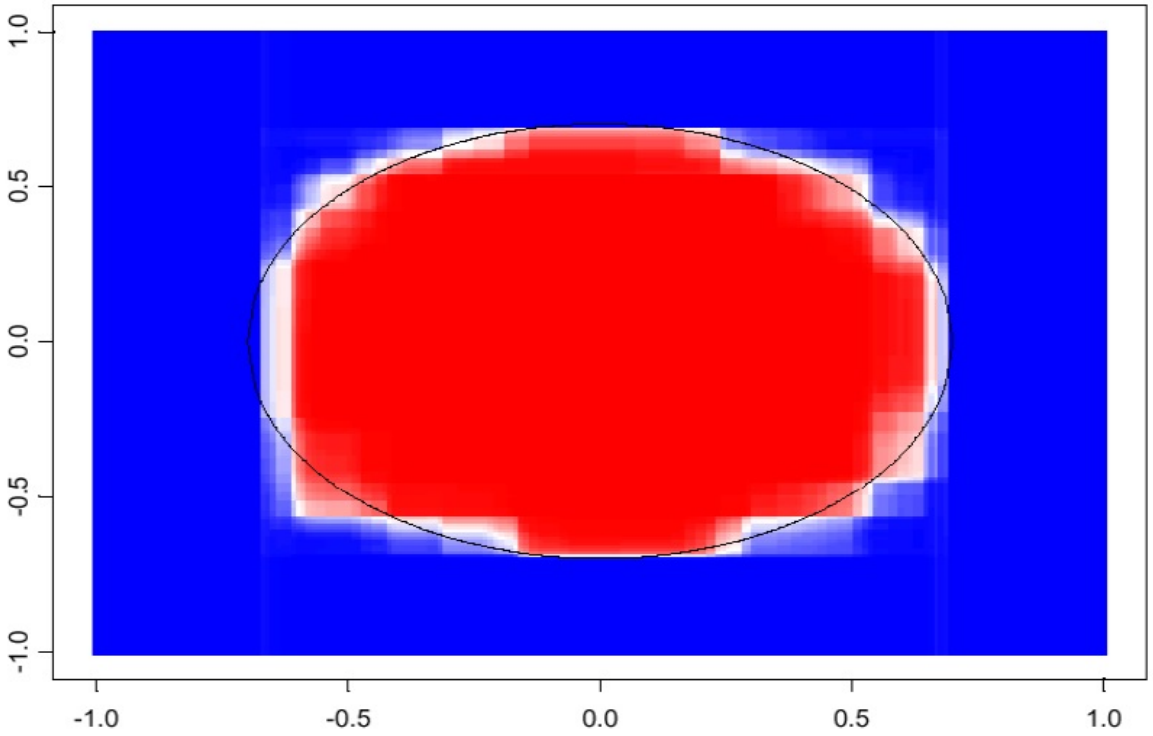
decision tree learning algorithm; very similar to ID3

CART decision boundary



Example

100 bagged trees



shades of blue/red indicate strength of vote for particular classification

Stacking

- Partition the data set into Training Tr, Validation V and Test Te sets
- Train different base classifiers $B_1, \ldots, B_k$ on the Tr
- Train a meta-classifier M using the predictions of the B_i on V
- Check performance of M on Te (usually better than the performance of the best B_i)[4]

[4] Dzoreski et al., "Is Combining Classifiers with Stacking Better ...," Machine Learning, 2004

The Ultimate Ensemble Algorithm?

Can we reduce both Bias and Variance?

- Bagging reduces variance by averaging
- Bagging has little effect on bias
- Can we average and reduce bias also?
- Yes:

Boosting

Adaboost

- Select a "weak learner" - an algorithm that tends to perform better than chance, but is not optimized for the problem.
- Results of applying a weak learner to training set is called "weak hypotheses" or "rules of thumb"
- Notation: Weak learner, $h : X \mapsto \{-1, 1\}$ – convenient to label the classes ± 1 rather than $0, 1$ – a training set $\{(x_1, y_1), \ldots, (x_n, y_n)\}$, and a distribution D: assigns a weight to each training point x_i – weight of $1/n$ is used initially
- The algorithm is iterative, it produces new weak learners and new distributions.

AdaBoost

1. Set $D_1(i) = 1/n$.
2. For t in $1, \ldots, T$:
   1. Let h_t be the weak learner trained on the data with weights D_t.
   2. Let ϵ_t be the error of h_t.
   3. Set $\alpha_t = \frac{1}{2} \log \left(\frac{1-\epsilon_t}{\epsilon_t} \right)$.
   4. Set
   $$D_{t+1}(i) = \frac{D_t(i) \exp(-\alpha_t y_i h_t(x_i))}{Z_t}$$
   where Z_t is the normalization required to make D_t sum to one.
3. Output $H(x) = \text{sign} \left(\sum_{t=1}^{T} \alpha_t h_t(x) \right)$.

AdaBoost

In D_{t+1}, if $h_t(x_i)$ is correct, the numerator is $\exp(-\alpha_t)$, and the weight is decreased, and if it is incorrect, the numerator is $\exp(\alpha_t)$ and the weight is increased.

Thus, Adaboost pushes later weak learners to focus more on the "hard" points, and less on the "easy" ones.

Reading Assignment

- Verma and Marchette, "Cybersecurity Analytics," Chapter 6
- Verma and Dyer, "On the Character of Phishing URLs: Accurate and Robust Statistical Learning Classifiers," ACM CODASPY 2015